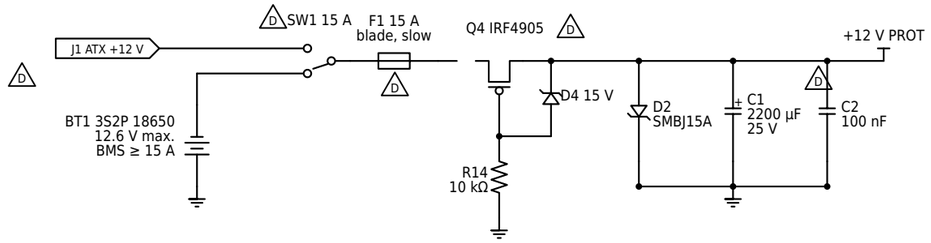
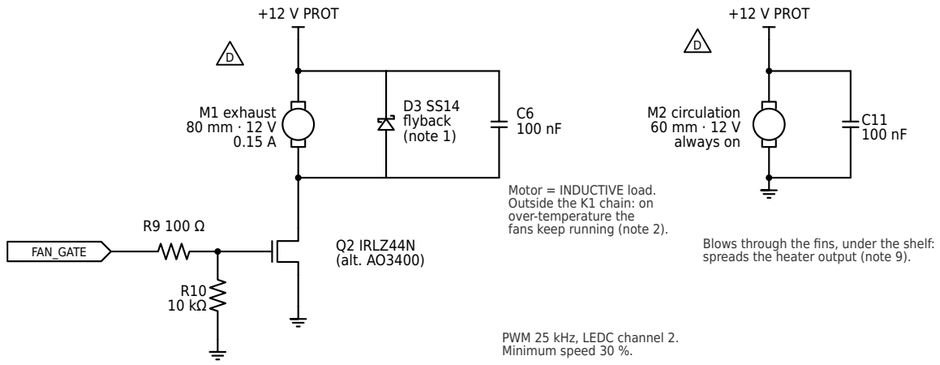
[1] INPUT · SOURCE SELECTION · PROTECTION



Q4: with correct polarity the MOSFET conducts (20 mΩ: ~1.7 W at 9 A, heatsinked). Reversed, it stays off. A Schottky would dissipate ~5 W at this current.

[7] VENTILATION · EXHAUST AND CIRCULATION

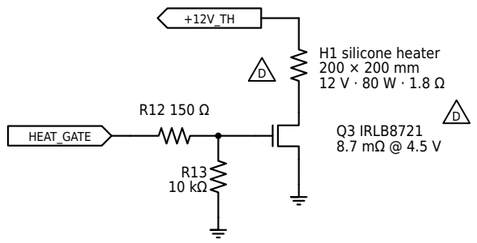


Motor = INDUCTIVE load. Outside the K1 chain: on over-temperature the fans keep running (note 2).

Blows through the fins, under the shelf: spreads the heater output (note 9).

PWM 25 kHz, LEDC channel 2. Minimum speed 30 %.

[9] BASE HEATER (LOW-SIDE)



RESISTIVE load: no flyback diode. Same command as the lamps (LEDC channel 1, same timer as 0). 6.7 A: Q3 dissipates ~0.4 W.

Heater bonded under two finned heatsinks; insulation below; perforated shelf above (note 9).

POWER NETS

NET	SOURCE	LOADS	CURRENT
+12 V PROT	J1 or BT1, via SW1, F1 and Q4	K1 and TH1/TH2 chain, M1, M2, U1, R2/R3	≈ 9.1 A max.
+12V_TH	K1 NO contact	EL1-EL5 (25 W), H1 (80 W)	≈ 8.8 A max.
+5 V	U1 (buck)	U2 (VIN), TS1	≈ 0.3 A peak
+3V3	U2 on-board regulator	DS1	≤ 30 mA

Supply: ATX ≥ 300 W with 12 V ≥ 12 A. The 3S2P battery holds 45 °C for ~1.5 h.

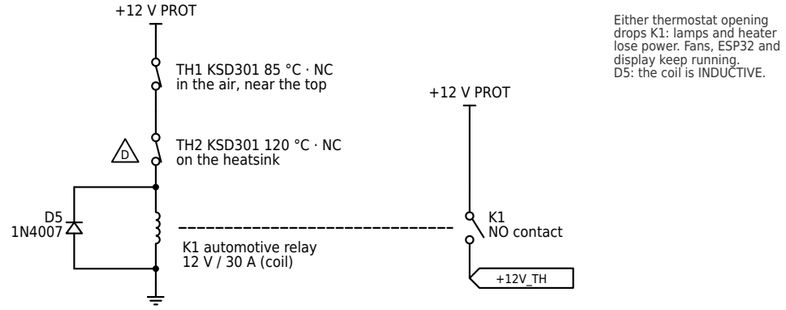
LEGEND

- D — change in this revision
- net label (connects by name)
- common ground
- junction

REVISIONS

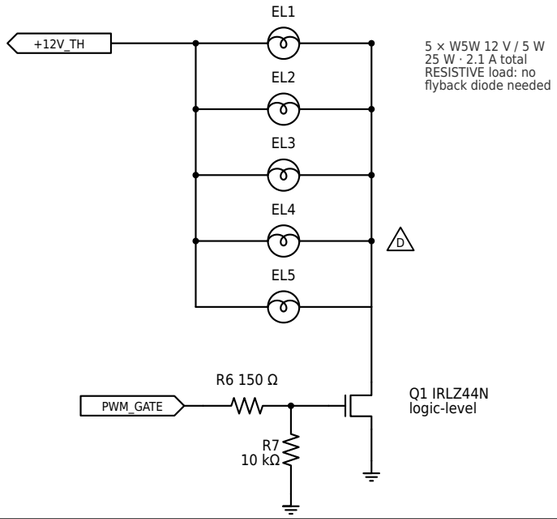
REV.	DESCRIPTION	DATE	BY
A	Initial release.	10/10/2026	GM
B	Fan M1 (Q2, D3, C6, R9, R10). Display DS1 TM1637 (C10). LM35DZ TS1 replaces DS18B20 (C8, R11, C9). ATX input = J1. Removed R1, R8, LED1.	11/10/2026	GM
C	Heater H1 30 W with Q3, R12, R13 and TH2. TH1 KSD301 10 A. F1 6.3 A, D1 SB560, SW1 10 A. GPIO27 = HEAT_GATE.	11/10/2026	GM
D	Polycarbonate chamber 25 cm (PIDL-MEC-001): 5 lamps, 80 W heater with Q3 IRLB8721, 80 mm fan and circulation fan M2 (C11). Safety chain TH1 85 °C + TH2 120 °C on the coil of relay K1 (D5), which feeds +12V_TH. Q4 IRF4905 replaces D1 (R14, D4). F1 15 A, SW1 15 A, C1 2200 µF, BT1 3S2P. Drawing split into 3 sheets.	11/10/2026	GM

[10] THERMAL SAFETY CHAIN

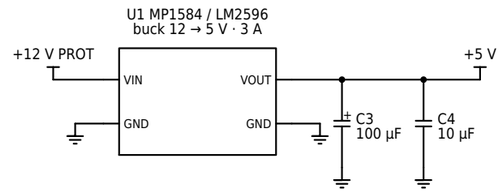


Either thermostat opening drops K1: lamps and heater lose power. Fans, ESP32 and display keep running.
D5: the coil is INDUCTIVE.

[6] LAMPS (LOW-SIDE)



[2] 12 V → 5 V REGULATION



SHEETS IN THIS DRAWING

1. Power: blocks [1] [2] [6] [7] [9] [10]
2. Control and sensing: blocks [3] [4] [5] [8]
3. Pinout, design notes and materials

WIRING

CIRCUIT	SIZE
Input, K1, +12V_TH, EL1-EL5, H1, Q1, Q3, Q4	1.5 mm²
M1, M2, K1 coil, TH1, TH2, U1	0.5 mm²
Signals, TS1 (twisted pair), DS1, +5 V	0.25 mm²

	NAME	INIT.	DATE
DRAWN	Gabriel Marques	GM	11/10/2026
APPROVED	Gabriel Marques	GM	11/10/2026

PROJECT PIDlampada

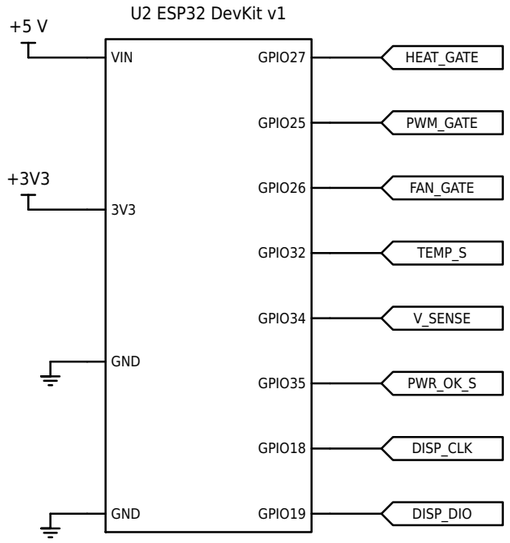
THERMAL CHAMBER WITH PWM CONTROL

ELECTRICAL SCHEMATIC · POWER

UNIP · COMPUTER ENGINEERING · CAPSTONE PROJECT 2026 — SÃO JOSÉ DOS CAMPOS - SP · BRAZIL

DRAWING No.	REV.	SHEET	SCALE	SIZE	DATE
PIDL-ESQ-001	D	1 / 3	NTS	A3	11/10/2026

[3] CONTROLLER

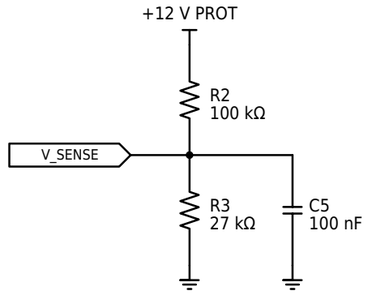


GPIO2: on-board status LED.
GPIO33 and GPIO13: reserved, not connected.

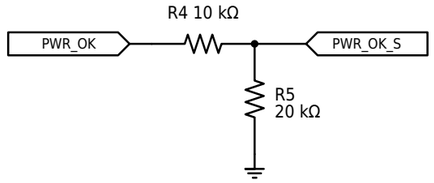
LEGEND

- D — change in this revision
- net label (connects by name)
- common ground
- junction

[5] ANALOG SENSING

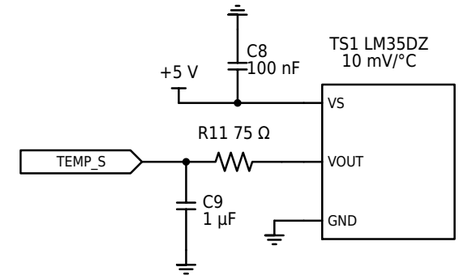


12.6 V → 2.68 V on ADC1.
ADC2 conflicts with Wi-Fi.



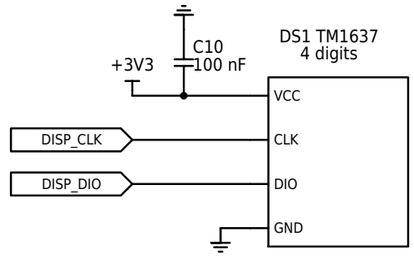
ATX PWR_OK: 5 V → 3.3 V.
No ATX reads 0 V (= battery).

[4] TEMPERATURE SENSOR



R11 + C9 filter per datasheet: the LM35 oscillates with a capacitor right at VOUT.

[8] TEMPERATURE DISPLAY



VCC at 3.3 V: the ESP32 does not tolerate 5 V on DIO (note 6).

GENERAL NOTES

1. Sheet 1: power. Sheet 2: control and sensing. Sheet 3: pinout, notes and materials.
2. Triangle with "D": added or changed in this revision. Earlier revisions in the table on sheet 1.
3. Net labels with the same name are connected, also across sheets.
4. All ground symbols are the same node: pack, ATX, U1 and U2 GND.
5. Resistors 1/4 W 5 %. Ceramic capacitors X7R 50 V. Unless noted.
6. Power loop (+12 V PROT, K1, +12V_TH, EL1-EL5, H1, Q1, Q3, Q4): 1.5 mm² wire (15 A).
7. Mechanical assembly, drilling and sheet cutting plan: drawing PIDL-MEC-001.

NET INTERCONNECTION

NET	SOURCE	DESTINATION	SHEETS
PWM_GATE	[3] U2 GPIO25	[6] Q1	2 → 1
HEAT_GATE	[3] U2 GPIO27	[9] Q3	2 → 1
FAN_GATE	[3] U2 GPIO26	[7] Q2	2 → 1
TEMP_S	[4] TS1 via R11/C9	[3] U2 GPIO32	2
V_SENSE	[5] R2/R3	[3] U2 GPIO34	2
PWR_OK_S	[5] R4/R5	[3] U2 GPIO35	2
DISP_CLK · DISP_DIO	[3] U2 GPIO18 · GPIO19	[8] DS1	2
+12 V PROT	[1]	[2] [7] [10] · [5]	1 → 1, 2
+12V_TH	[10] contato do K1	[6] [9]	1
+5 V	[2] U1	[3] U2 · [4] TS1	1 → 2
+3V3	[3] U2	[8] DS1	2

	NAME	INIT.	DATE
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PROJECT PIDlampada

THERMAL CHAMBER WITH PWM CONTROL

ELECTRICAL SCHEMATIC · CONTROL AND SENSING

UNIP · COMPUTER ENGINEERING · CAPSTONE PROJECT 2026 — SÃO JOSÉ DOS CAMPOS - SP · BRAZIL

DRAWING No.	REV.	SHEET	SCALE	SIZE	DATE
PIDL-ESQ-001	D	2 / 3	NTS	A3	11/10/2026

PINOUT · U2 ESP32 DEVKIT V1

PIN	NET	TYPE	FUNCTION	REV.
GPIO25	PWM_GATE	output	Q1 gate (5 lamps) · LEDC channel 0, 500 Hz, 10 bit	A
GPIO27	HEAT_GATE	output	Q3 gate (80 W heater) · LEDC channel 1, same timer as the lamps	C
GPIO26	FAN_GATE	output	Q2 gate (80 mm fan) · LEDC channel 2, 25 kHz, 8 bit	B
GPIO32	TEMP_S	analog	ADC1_CH4 ← LM35 through R11/C9 filter	B
GPIO34	V_SENSE	analog	ADC1_CH6 ← R2/R3 divider (bus voltage)	A
GPIO35	PWR_OK_S	analog	ADC1_CH7 ← R4/R5 divider (ATX PWR_OK)	A
GPIO18	DISP_CLK	output	TM1637 clock (DS1)	B
GPIO19	DISP_DIO	bidir.	TM1637 data, open drain, pull-up on module	B
GPIO2	–	output	On-board status LED	B
GPIO33	SETPOINT	analog	Reserved: setpoint potentiometer (simulation only)	A
GPIO13	BTN_MODE	input	Reserved: mode button	C
VIN	+5 V	supply	5 V input from U1	A
3V3	+3V3	supply	On-board regulator output → DS1	A
GND	GND	ground	Common ground	A

DESIGN NOTES

1. **Flyback diodes only on inductive loads.** Lamps and heater are resistive and need no diode. The fan motor (D3) and the coil of relay K1 (D5) are inductive: at turn-off the winding current needs a path and, without the diode, becomes a voltage spike that kills the MOSFET or the thermostat. D3 is a Schottky because the PWM runs at 25 kHz.
2. **Fans outside the safety chain.** On over-temperature K1 cuts lamps and heater and the firmware drives the fan at 100 %. If the fans were cut too, the chamber would only cool by convection.
3. **Split-range control.** A single PI with output from −1 to 1: positive drives lamps and heater, negative drives the fan, never both. The fan has a 30 % minimum speed and on/off hysteresis.
4. **LEDC channels.** Channels come in pairs sharing one timer (0-1, 2-3…). Lamps (0) and heater (1) share the 500 Hz timer on purpose; the fan uses channel 2 so its 25 kHz does not change that frequency.
5. **LM35 at 5 V.** The LM35 needs 4 to 30 V. Its output, 0.25 to 0.70 V over the useful range, goes to the ADC with no divider. The LM35DZ reads 0 to 100 °C: the 70 °C setpoint ceiling leaves margin.
6. **TM1637 at 3.3 V.** The module pull-ups go to VCC. With the module at 5 V the ESP32 DIO pin would see more than its limit.
7. **Input protection.** Q4 (P-MOSFET) is the reverse-polarity protection: it conducts with 20 mΩ when polarity is right and blocks when reversed. D4 clamps VGS. D2 absorbs hot-plug transients; F1 is slow for the inrush of cold lamps and C1. R7, R10 and R13 hold the gates at 0 V during boot.
8. **Voltage compensation.** On battery the bus drops from 12.6 to ~10 V. The firmware scales the lamp and heater duty by $(12/V)^2$ and the fan duty by 12/V, reading V on V_SENSE.
9. **Heater, heatsinks and shelf.** H1 is bonded under two finned heatsinks, on ceramic-fibre insulation. Fan M2, always on, blows air through the fins; the warm air rises through the holes of the perforated aluminium shelf that holds the samples. Lamps and heater get the same command; the feedforward uses the combined gain (16 + 50 °C).
10. **Safety chain.** TH1 (85 °C, in the air) and TH2 (120 °C, on the heatsink) in series with the coil of K1. Either one opening drops the relay and removes power from lamps and heater; the firmware also cuts everything above 78 °C. The thermostats only carry the ~80 mA coil current, not the 8.8 A load.
11. **Polycarbonate chamber.** 6 mm clear solid PC (U ≈ 3.9 W/m²K) withstands continuous use up to ~115 °C. 25 × 25 × 25 cm chamber: UA ≈ 1.6 W/K and time constant ≈ 1000 s, which is why power went up to 105 W. See PIDL-MEC-001.

BILL OF MATERIALS (ELECTRICAL)

ITEM	REF.	QTY	DESCRIPTION	RATING	REV.
1	**BT1**	1	3S2P 18650 pack (high drain) with BMS	12.6 V · BMS ≥ 15 A	△D
2	**J1**	1	ATX connector (4-pin Molex), PSU ≥ 300 W	12 V ≥ 12 A	△D
3	**SW1**	1	SPDT selector switch	15 A · ATX / battery	△D
4	**F1**	1	Blade fuse + holder	15 A slow	△D
5	**Q4**	1	IRF4905 P-channel MOSFET + heatsink	55 V · 20 mΩ	△D
6	**D4**	1	Zener diode	15 V · 0.5 W	△D
7	**D2**	1	SMBJ15A TVS suppressor	15 V · 600 W	A
8	**C1**	1	Electrolytic capacitor	2200 µF · 25 V	△D
9	**C2, C5, C6, C8, C10, C11**	6	Ceramic capacitor	100 nF · 50 V	△D
10	**U1**	1	MP1584 / LM2596 buck module	12 → 5 V · 3 A	A
11	**C3 · C4**	2	Electrolytic · ceramic	100 µF · 10 µF, 16 V	A
12	**U2**	1	ESP32 DevKit v1 board	30-pin	A
13	**TH1**	1	KSD301 bimetal thermostat (in the air)	85 °C · NC	△D
14	**TH2**	1	KSD301 bimetal thermostat (on heatsink)	120 °C · NC	△D
15	**K1**	1	Automotive relay + socket	12 V · 30 A · NO	△D
16	**D5**	1	1N4007 rectifier diode	1 A · 1000 V	△D
17	**EL1-EL5**	5	W5W halogen lamp with T10 socket	12 V · 5 W	△D
18	**Q1, Q2**	2	IRLZ44N N-channel MOSFET (Q2 alt. AO3400)	logic-level	A
19	**Q3**	1	IRLB8721 N-channel MOSFET	8.7 mΩ @ 4.5 V	△D
20	**H1**	1	200 × 200 mm silicone heater pad	12 V · 80 W · 1.8 Ω	△D
21	**M1**	1	80 × 80 × 25 mm exhaust fan	12 V · 0.15 A	△D
22	**M2**	1	60 × 60 × 15 mm circulation fan	12 V · 0.1 A	△D
23	**D3**	1	SS14 Schottky diode (alt. 1N5819)	1 A · 40 V	B
24	**TS1**	1	LM35DZ temperature sensor	10 mV/°C · TO-92	B
25	**C9**	1	Ceramic capacitor	1 µF · 16 V	B
26	**DS1**	1	TM1637 display module	4 digits with point	B
27	**R2 · R3 · R5**	3	Resistor	100 kΩ · 27 kΩ · 20 kΩ	A
28	**R4, R7, R10, R13, R14**	5	Resistor	10 kΩ	△D
29	**R6, R12 · R9 · R11**	4	Resistor	150 Ω · 100 Ω · 75 Ω	C

Mechanical parts (PC sheet, angles, MDF base, hinges): list on PIDL-MEC-001. Removed over the revisions: R1, R8, LED1, DS18B20 (B) and D1 (D, replaced by Q4).

	NAME	INIT.	DATE
DRAWN	Gabriel Marques	GM	11/10/2026
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PROJECT PIDlampada

THERMAL CHAMBER WITH PWM CONTROL

PINOUT · DESIGN NOTES · BILL OF MATERIALS

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DRAWING No.	REV.	SHEET	SCALE	SIZE	DATE
PIDL-ESQ-001	**D**	3 / 3	NTS	A3	11/10/2026